Research Plan: Lamb Waves in a Piezoelectric Quasicrystal Plate

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Problem statement

Consider a thin plate of thickness 2h made of a one-dimensional hexagonal piezoelectric quasicrystal. Both faces of the plate are free. Besides the usual elastic (phonon) field and the electric potential, the material has a phason field. Heat conduction follows the Moore–Gibson–Thompson (MGT) model, and size effects are described by the nonlocal strain gradient theory, which has two length scales.

Study the Lamb waves that travel along this plate. For the symmetric and antisymmetric modes, find the phase velocity and the attenuation as functions of frequency and plate thickness. Show how they change with the phonon–phason coupling, the nonlocal length, the gradient length and the thermal relaxation time, and compare open-circuit and short-circuit faces.

Understanding the problem

This section explains every term used in the problem statement, one by one. If a word in a paper looks unfamiliar later, come back here first.

Why a plate? Many wave papers treat a half-space, a block of material that goes on forever below its surface. Real devices use thin plates or films, and a plate behaves differently because the wave keeps bouncing between its two faces. Why a quasicrystal? It is an unusual kind of solid with an extra field inside it, and that extra field changes how waves travel.

Wave and plate

A wave is a disturbance that moves through a material and carries energy with it. Shake one end of a long rope and the bump runs along the rope, while the rope itself stays where it is. A plate is a flat body whose thickness is much smaller than its length and width, like a steel ruler or a sheet of tin.

Lamb wave

In a plate, a wave keeps reflecting from the top and bottom faces. The reflected parts add up and form a pattern that moves along the plate. This guided wave is called a Lamb wave, after Horace Lamb, who described it in 1917. The ringing you hear when you tap a thin metal sheet is the whole sheet vibrating, not only its surface.

Frequency, wavenumber and speed

The angular frequency ω is 2π times the number of vibrations per second. The wavenumber k is 2π divided by the wavelength. The phase velocity is $c = \omega/k$, which is the speed at which a crest moves along the plate.

Dispersion

In a plate, waves of different frequencies travel at different speeds. This is called dispersion. A glass prism shows the same idea with light: each colour moves at a slightly different speed in glass, so white light spreads into a rainbow. A dispersion curve is a graph of frequency against wavenumber, and most of our figures are built from it.

Modes

At a given frequency, a plate can carry a wave in more than one pattern, and each pattern is called a mode. In the usual case, symmetric modes stretch and squeeze the plate along its length, and antisymmetric modes bend it like a flapping sheet. They are named S0, S1, S2 and so on, and A0, A1, A2 and so on. In a piezoelectric quasicrystal plate, whether the modes split cleanly into these two families depends on the direction of the quasiperiodic axis and the polarisation. We will fix those directions in step 2.

Cutoff frequency

Higher modes do not exist at every frequency. Each one appears only above its own cutoff frequency. A road is a rough picture: a narrow road carries one line of vehicles and a wider road carries several. At low frequency the plate carries only the first modes, and more become possible as the frequency rises.

Quasicrystal

In an ordinary crystal, atoms sit in a pattern that repeats, like the tiles on a bathroom floor: shift the pattern by one tile and it looks the same. A quasicrystal is ordered too, but its pattern never repeats exactly. The best picture is a Penrose tiling, a floor made of two

kinds of tile that fills the space with no gaps and looks neatly ordered, yet no shift ever makes it match itself. Dan Shechtman first observed such a solid in the early 1980s, and he received the 2011 Nobel Prize in Chemistry for it. Quasicrystals are usually hard and brittle, and they conduct heat poorly.

We work with a one-dimensional (1D) hexagonal quasicrystal. Its atoms follow the non-repeating pattern along one direction only, and in the plane perpendicular to it they repeat as in a normal crystal. This type is used in most published wave papers, which helps us check our results.

Phonon field and phason field

A normal crystal has one kind of atomic motion, the ordinary displacement of atoms when a wave passes. In a quasicrystal this is called the phonon field and is written u. A quasicrystal has a second kind of motion, the phason field, written w. In the tiling picture, a phason is a small local rearrangement, such as a few tiles flipping into another arrangement, that leaves the overall look unchanged. A phonon shifts or stretches the pattern, and a phason rearranges it.

The two fields push on each other. This phonon–phason coupling is controlled by coupling constants, and it is the main new ingredient compared with an ordinary plate. How the phason moves, as a wave or as a slow diffusion, differs between papers. We will decide this in step 3 after reading them.

Piezoelectricity

Some crystals produce a voltage when they are squeezed, and they change shape when a voltage is applied to them. The spark in a gas lighter comes from pressing such a crystal, and a quartz watch uses the opposite effect. In a piezoelectric quasicrystal the phonon and phason fields are both linked to the electric potential. Experimental data on real piezoelectric quasicrystals is limited, so many papers take material constants from a few sources or assume them, and we will say so openly.

Heat and the MGT model

Stretching a material slightly changes its temperature, and a change in temperature makes it expand or shrink. Stretch a rubber band quickly and touch it to your lip, and it feels warm. So elastic waves and heat are linked, and that is why the problem is called thermoelastic.

The classical law of heat conduction, Fourier's law, says that a temperature change is felt everywhere at once. That cannot be exactly true. Put a pan on a stove and the handle takes a while to warm up. The Moore–Gibson–Thompson (MGT) model fixes this by letting heat travel at a finite speed, using a thermal relaxation time.

Size effect

Classical elasticity says that the stress at a point depends only on the strain at that same point. That works very well for everyday objects. At micro and nano scales, tests on tiny beams and plates show behaviour that this rule cannot explain.

Nonlocal theory, started by Eringen, says the stress at a point depends on the strain in a neighbourhood around it. Strain gradient theory says it also depends on how quickly the strain changes from place to place. The nonlocal strain gradient theory uses both ideas and has two length scales, one for each. When the plate is much thicker than these lengths, the classical answer comes back. Think of a person in a crowded lift: what they feel depends on the people around them, not only on the spot where they stand.

Attenuation and complex wavenumber

If a wave loses energy as it travels, the wavenumber is no longer a real number. It is written k = k_r + i k_i. The real part k_r gives the wavelength and the speed. The imaginary part k_i tells how fast the amplitude falls, in proportion to e^(−k_i x). A struck bell is the usual picture: the sound is loud at first and then slowly fades. In our plate, heat flow is one cause of fading, and slow phason motion may be another, depending on how we model it. When the figures mention complex branches, they mean curves where k has both a real and an imaginary part.

The model we will use

The model is kept small so that each part can be checked on its own.

The plate has thickness 2h, and the wave travels along its length. We work in plane strain, which means the motion takes place in the length and thickness directions only. The quasiperiodic axis is taken along the thickness, as in most published plate papers; we confirm this choice in step 2 while reading them. The top and bottom faces carry no stress, phonon or phason. They can be thermally insulated or held at a fixed temperature, and electrically open or short-circuited.

Inside the plate we use the phonon and phason equations with piezoelectric coupling, the electric equation in its quasi-static form, and the MGT heat equation. The size effect enters through the nonlocal strain gradient law with two length scales. We first apply it to the phonon stresses. Whether to apply it to the phason stresses as well is decided in step 5.

We leave out surface elasticity, flexoelectricity, memory-dependent derivatives, imperfect interfaces, layered or graded plates, nonlinear effects and full 3D geometry. Each of these can be a separate paper later.

Two points must be written clearly in the paper. First, the size-effect theory asks for extra boundary conditions on the faces. We apply the usual ones, say so openly, and show how much the results change when the length scales change. Second, material constants for piezoelectric quasicrystals are not well measured. We use ranges taken from the literature, never call a value measured, and show how sensitive the results are to each constant.

Work plan

The work is done in six steps. After each step there is a check against a known result. Please send me that check plot before you start the next step, so that if something is wrong we know which step caused it.

Step	What you do	The check before moving on	What you gain
1. Plain elastic plate	Derive and solve the classical Rayleigh–Lamb problem for the phonon field only	Known cutoff frequencies and thin-plate limits match; for a very thick plate the speed approaches the Rayleigh half-space speed	Confidence with the method
2. Add piezoelectricity	Include the electric potential, for open and short-circuited faces	Switching the coupling off returns step 1; the open/short speed difference gives the coupling coefficient K^2	A first result worth showing

Step	What you do	The check before moving on	What you gain
3. Add the phason field	Include the phason equation and the phonon–phason coupling	Switching the phason coupling off returns step 2; the dispersion curves agree with a published 1D hexagonal piezoelectric quasicrystal plate result from the reading list	The quasicrystal part of the paper
4. Add heat (MGT)	Include the temperature field and the MGT heat equation	Switching thermal coupling off returns step 3; the classical thermoelastic limit agrees with published results	A thermal effect on speed and fading
5. Add the size effect	Include the nonlocal strain gradient terms	Setting both length scales to zero returns step 4; nonlocal-only and gradient-only cases are checked separately	The main new contribution
6. Analysis	Track complex branches, fading, maps and sensitivity	Branches vary smoothly and root counts stay consistent along each branch	The figures that carry the paper

After step 4 we already have a verified thermal piezoelectric quasicrystal plate, which is a usable base for the paper. Step 5 adds the new part. If step 5 turns out to be heavy, we will decide together what to do.

Calculations and figures

For each wavenumber we find the complex frequencies at which the plate can carry a wave, and then follow each branch from one wavenumber to the next so that curves do not get mixed up. I will give you working code for this part, so your time goes into

understanding the results and making the figures.

Figure	What it shows	Why the paper needs it
Dispersion curves	Frequency against wavenumber for each mode	The base result
Complex branches	Real and imaginary parts together	Shows the full wave behaviour, not only the real curves
Phase speed against frequency–thickness	Where each mode switches on and how it settles	Shows true plate behaviour
Phonon–phason coupling maps	How the coupling constants change speed and mode order	The quasicrystal result of the paper
Fading against frequency	Effect of heat, and of phason motion if it is modelled as slow, on how quickly waves die out	Links to device performance
Thickness–frequency–speed surface	The whole parameter landscape in one view	Rich information in one figure
Length-scale contour maps	Separate effects of the two size parameters	The main result of the size model
Coupling coefficient K^2 against thickness	Open versus short-circuit difference	Useful for sensor design
Mode shapes	Phonon displacement, phason displacement, potential and temperature across the thickness	Builds physical understanding
Sensitivity ranking	Which constant changes the speed most	Shows how far results can be trusted
Verification table	Each limiting case and its match	Builds reviewers′ trust

All figures use one colour scheme, one font size and a resolution of 300 dpi, so the paper looks like one consistent piece of work.

Possible difficulties

These are the problems I expect along the way. None of them is unusual.

Difficulty	What it looks like	How we handle it
Closely related papers already exist	Lamb waves in piezoelectric quasicrystal plates, in thermoelastic quasicrystal nanoplates and with nonlocal theory have each been published separately	Read these papers fully first, state exactly which combination is new, and do not claim more
Extra boundary conditions in the size-effect theory	A reviewer may ask why only the usual conditions are applied	We state our choice openly and show how little or how much the results move when the length scales change
How to model the phason	Papers treat it differently, as wave-like or as slow diffusion	Choose after reading, say why, and show that the main conclusions survive the other choice if the time allows
Uncertain material constants	Piezoelectric quasicrystal data is limited and varies between sources	Use literature ranges, never call a value measured, and show the sensitivity
A missing or tangled branch	A curve jumps or two curves swap places	Follow each branch step by step and check the root count along the way
Too much at once	The full model can feel large	Work step by step, and treat step 5 as the last addition

If you get stuck at any step, send me the plot that looks wrong and we will go through it together.

Reading list

Nine items are enough to start. Read the first two first, since they show how a piezoelectric quasicrystal plate is set up, then the thermal one in item 5. Read for the idea first and the algebra afterwards.

#	Paper	Why read it
1	Guided wave propagating in a 1-D hexagonal piezoelectric quasi-crystal plate, Acta Mechanica, 2020	The basic plate model with phonon, phason and electric fields; a source for checking step 3
2	He, Lamb Waves in One-Dimensional Hexagonal Piezoelectric Quasicrystal Layered Plates, Academic Journal of Applied Sciences, 2026	A recent Lamb wave study with coupled mechanical, phason and electric fields, solved numerically
3	Mode conversions and intersections of Lamb waves in 1D hexagonal piezoelectric quasicrystal nanoplates based on the integral nonlocal theory	A size-dependent quasicrystal plate paper; compare its nonlocal treatment with ours
4	Lamb waves in piezoelectric quasicrystal multi-layered nano-plates with imperfect interfaces, 2025	A recent paper on the same material; shows what is already covered
5	Generalized Thermoelastic Lamb Waves in One-Dimensional Hexagonal Quasicrystal Nanoplates, 2026	The closest thermal quasicrystal plate paper; we compare our heat model against it
6	Ray, Dispersion curves for Rayleigh–Lamb waves in a micro-plate considering strain gradient elasticity, Wave Motion 86, 2019	How a strain gradient plate is set up and how its dispersion is built
7	Lim, Zhang and Reddy, A higher-order nonlocal elasticity and strain gradient theory and its applications in wave propagation, J. Mech. Phys. Solids 78, 2015	The source of the nonlocal strain gradient theory; confirm the exact citation before citing
8	Shaat, Infeasibility of the nonlocal strain gradient theory for applied Physics, arXiv, 2017	A critique of the theory; good to know so we can answer it if a reviewer raises it

#	Paper	Why read it
9	Our own MGT papers, for example Gupta et al., Piezo-semiconductor-void interplay in the size-dependent dynamics of Rayleigh waves within the Moore–Gibson–Thompson framework, IJNMHFF 35(9), 2025	The MGT heat equation and its coupling that we will reuse

Outline of the paper

1. **Introduction:** why plates matter, what quasicrystals add, what size-dependent theories add beyond the classical Eringen model, and what the closest earlier papers did and did not cover.
2. **Formulation:** the plate, the field equations for the phonon, phason, electric and MGT heat fields, the size-effect law, and the boundary conditions with the clear note on extra conditions.
3. **Dispersion relation:** the determinant, and how the modes split into families, depending on the quasiperiodic axis and the polarisation.
4. **Numerical method and verification:** how roots are found and followed, and the table of limiting cases.
5. **Results:** dispersion and complex branches, phonon–phason coupling maps, fading, the thickness surface, length-scale maps, K^2, mode shapes and sensitivity.
6. **Conclusions:** what each claim showed, the limitations, and future work such as surface elasticity, imperfect interfaces and graded plates.
7. **Supplementary material:** the code and the full parameter table, so anyone can reproduce the results.

The claims to keep in view while working, to be finalised after the reading of items 1 to 5 above:

- A coupled piezoelectric quasicrystal plate model with MGT heat and a two-length-scale size effect.
- Real and complex branches followed together, with fading and mode behaviour.
- Every step verified against a known limit.